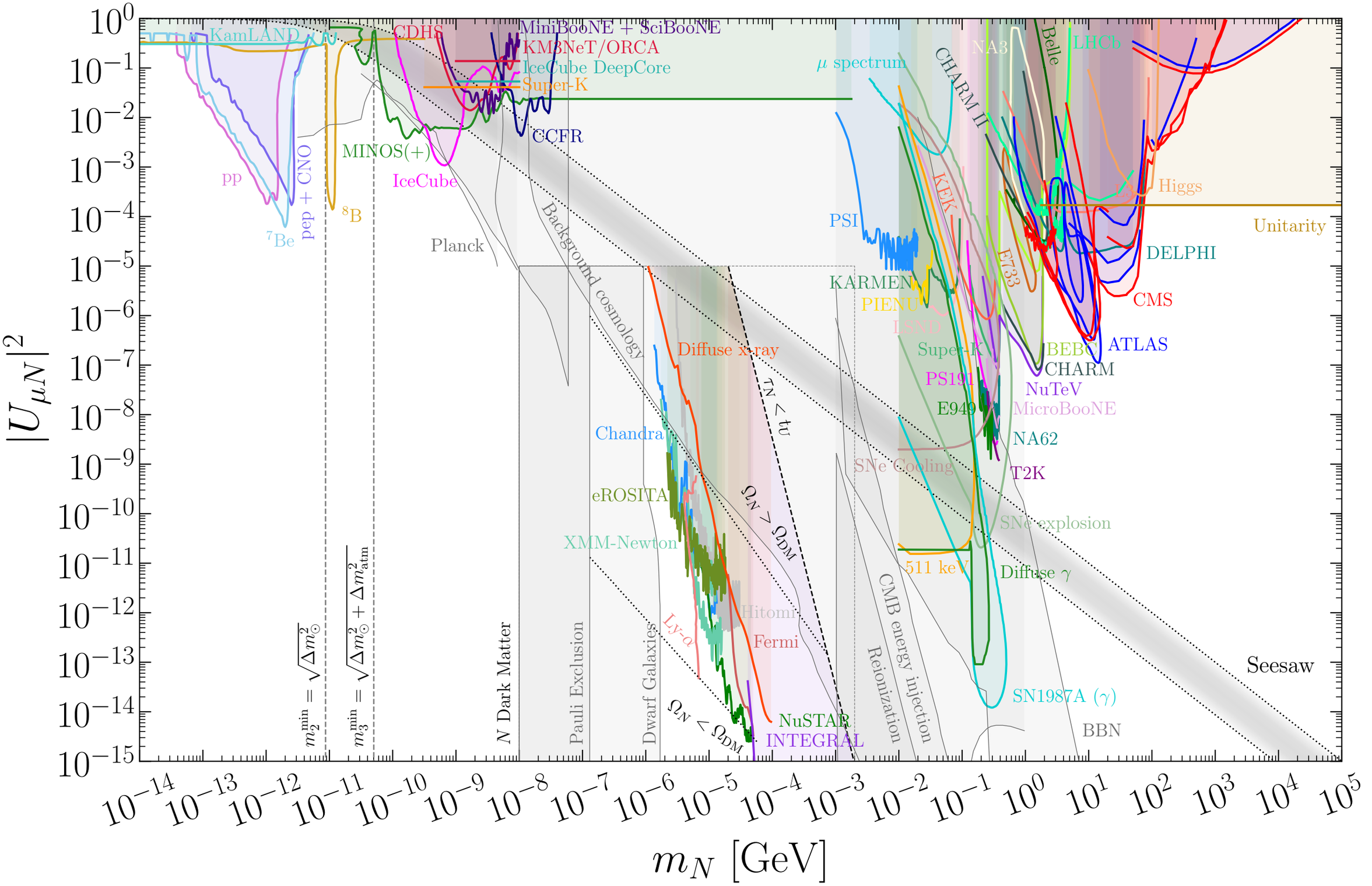

$|U_{\mu N}|^2$
m_N [GeV]
KamLAND
CDHS
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
μ spectrum
CHARM II
NA3
Belle
LHCb
pp
^{7}Be
pep + CNO
^{8}B
MINOS(+)
IceCube
CCFR
Planck
Background cosmology
PSI
KEK
E733
Higgs
Unitarity
DELPHI
CMS
ATLAS
KARMEN
PIENU
LSND
Super-K
BEBC
CHARM
PS191
NuTeV
E949
MicroBooNE
NA62
T2K
SNe Cooling
SNe explosion
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
XMM-Newton
$\Omega_N > \Omega_{\rm DM}$
511 keV
Diffuse γ
Hitomi
Ly-α
Fermi
$\Omega_N < \Omega_{\rm DM}$
NuSTAR
INTEGRAL
SN1987A (γ)
BBN
Seesaw
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB energy injection
Reionization